The Classical P vs NP Problem is Mathematically and Physically Ill-Posed

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Abstract

This work establishes a physically–meaningful separation between **P** and **NP** by embedding computation within the curvature constraints of Wave Confinement Theory (WCT).

We introduce the Wave Curvature Computation (WCC) model, in which information is stored as spatially localized soliton states whose stability and distinguishability require finite geometric curvature. We prove an *internal* separation $\mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}}$ by showing that the Locked–Expander 3-SAT family has witness configurations whose curvature energy is $2^{\Omega(n)}$, while every physically realizable WCC evolution is limited by a polynomial curvature–capacity law derived from QFT regularity, locality bounds, and gravitational backreaction.

Next, we show that the discrete WCC model implemented in finite precision (FP32/FP64) satisfies the three uniformity criteria for Turing simulation— polynomial-time encodability, constant-size operator descriptions, and polynomial-time decodability—yielding $\mathbf{P}_{\text{WCC}} = \mathbf{P}_{\text{Turing}}$.

Combining these results, we obtain a *physical* separation:

$$\mathbf{P}_{\text{physical}} = \mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}} = \mathbf{NP}_{\text{physical}}.$$

This demonstrates that NP witnesses for the Locked–Expander family cannot be constructed in any finite-energy, finite-precision, curvature-bounded physical universe. The classical question $P \stackrel{?}{=} NP$ for idealized curvature-free Turing Machines is thereby identified as a metaphysical abstraction, not a question about physically realizable computation.

1 Historical Context: The Divergence of Logic and Physics

Alan Turing's 1936 paper introduced the Turing Machine as a model of *effective calculability*: a formal mechanism for deciding what can be computed in principle. The machine was intentionally defined as a *logical abstraction*, with an infinite tape and unit-cost access to arbitrarily distant memory cells. No claim was made that the infinite tape constitutes a physically realistic model of efficient computation.

The identification of "Turing Machine steps" with *physical time* is a much later development, codified in the *Extended Church–Turing Thesis (ECT)*, which asserts:

> Any physically realizable computation can be simulated by a Turing Machine with at most polynomial overhead.

This assumption silently removes the physical cost of maintaining and manipulating information. In a classical Turing Machine, accessing the n-th memory cell costs unit time, regardless of the physical distance, energy density, or stabilizing curvature required to store that bit in a real medium. The model presumes curvature-free memory, zero erasure cost, and unbounded, thermodynamically neutral precision.

In contrast, physically realizable information in standard spacetime obeys geometric constraints: wave confinement, locality bounds, Landauer's principle, finite-energy storage, and curvature backreaction. Real computation necessarily has a curvature budget.

This work argues that the ECT is physically incorrect. The classical P vs. NP problem thereby concerns the behavior of a mathematically convenient but physically impossible device. By imposing the Curvature Capacity Rail, the WCC model restores the connection between computation and physics, producing a physically meaningful separation:

$$\mathbf{P}_{\text{physical}} \neq \mathbf{NP}_{\text{physical}}.$$

2 The Physical Distinguishability Obstruction

Classical complexity theory treats the alphabet $\{0, 1\}$ as consisting of two perfectly distinguishable symbols. A Turing machine is allowed to read and write 0 or 1 on any tape cell with no physical cost, and without introducing any geometric, energetic, or structural change to the machine. This section establishes that such a model is not physically realizable: distinguishability of logical states requires nonzero curvature.

2.1 Physical Encoding of Logical Bits

A logical state $b \in \{0, 1\}$ must be represented by a physical field configuration

$$\psi_b : \Omega \subset \mathbb{R}^3 \to \mathbb{C}.$$

For two symbols to be reliably distinguishable, the corresponding physical states must satisfy

$$\|\psi_0 - \psi_1\|_{H^1(\Omega)} > 0.$$

Otherwise, noise renders them indistinguishable.

Definition 2.1 (Curvature Cost of Distinguishability). *For two physical encodings ψ_0 and ψ_1, define the curvature separation cost*

$$\Delta C_\Theta(\psi_0, \psi_1) := C_\Theta(\psi_1 - \psi_0) = \int_\Omega \left| \frac{\Delta(\psi_1 - \psi_0)}{(\psi_1 - \psi_0) + \epsilon} \right|^2 dx.$$

This quantity measures the geometric energy required to implement a discrete change of logical value.

Lemma 2.2 (Distinguishability Requires Curvature). *If two logical symbols 0 and 1 are encoded in physically distinct states, then*

$$\Delta C_\Theta(\psi_0, \psi_1) > 0.$$

Proof. If $\psi_0 \neq \psi_1$ in $H^1(\Omega)$, then $\Delta(\psi_1 - \psi_0)$ is not identically zero. Because $|\psi_1 - \psi_0 + \epsilon| \geq \epsilon$, the integrand defining C_Θ is strictly positive on a set of nonzero measure, implying $\Delta C_\Theta > 0$. □

2.2 Curvature Cost of an n-bit Configuration

Let a witness string $w = (b_1, \dots, b_n) \in \{0, 1\}^n$ be physically encoded as a set of spatially separated field regions

$$\psi_w = \psi_{b_1} \oplus \cdots \oplus \psi_{b_n}.$$

Distinct memory locations require distinct spatial regions; therefore the curvature of the full configuration is at least the sum of the curvature cost of the individual distinctions:

$$C_\Theta(\psi_w) \;\geq\; \sum_{i=1}^{n} \Delta C_\Theta(\psi_0, \psi_1) \;\asymp\; n \cdot c_0,$$

for some fixed $c_0 > 0$ determined by the physical representation of a bit.

Proposition 2.3 (Bit Storage Requires Linear Curvature). *Every physically encoded n-bit configuration satisfies*

$$C_\Theta(\psi_w) \geq c_0 n.$$

Thus, the very existence of an n-bit witness in physical memory has a nonzero geometric cost.

2.3 Classical Turing Machines Ignore Curvature

In the Turing model, a witness

$$w = (b_1, \dots, b_n) \in \{0, 1\}^n$$

is stored on tape with the following idealizations:

- all memory cells are distinguishable with zero physical structure;
- writing 0 or 1 has no physical cost;
- there is no spatial embedding of information;
- curvature, energy, gradient, and entropy cost are all ignored.

Formally, one assumes:

$$C_\Theta(\psi_0) = 0, \qquad C_\Theta(\psi_1) = 0, \qquad C_\Theta(\psi_w) = 0.$$

This violates Lemma 2.2 and Proposition 2.3. Therefore the classical Turing model does not correspond to any physically realizable computational architecture.

Corollary 2.4 (The Turing Model Is Not Physically Realizable)**.** *No finite-energy, finite-radius physical system can satisfy the assumptions of the classical Turing model regarding cost-free distinguishability of symbols. Hence*

$$\mathbf{P}_{Turing} \quad \textit{overestimates actual physical computational power.}$$

2.4 Physical Complexity Classes

We therefore define the physically meaningful analogues:

$$\mathbf{P}_{\text{physical}} := \mathbf{P}_{\text{WCC}}, \qquad \mathbf{NP}_{\text{physical}} := \mathbf{NP}_{\text{WCC}}.$$

Using the curvature-capacity theorem,

$$\mathbf{P}_{\text{WCC}} = \mathbf{P}, \qquad \mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}},$$

thus yields the *physical* separation:

$$\mathbf{P}_{\text{physical}} \neq \mathbf{NP}_{\text{physical}}.$$

Theorem 2.5 (Physical Separation of **P** and **NP**)**.** *If computation is constrained by finite curvature capacity, then*

$$\mathbf{P} \neq \mathbf{NP} \quad \textit{in physically realizable systems.}$$

3 The Classical Turing Model Is Not a Physical or Mathematical Model

In this section we address a foundational issue: the classical Turing Machine (TM) is neither a physical model of computation nor a fully axiomatized mathematical structure. Its operational assumptions— infinite tape, perfect symbols, cost-free erasure, and geometry-free information—constitute *metaphysical idealizations.* These idealizations invalidate the inference

$$\mathbf{P}_{\text{Turing}} = \mathbf{NP}_{\text{Turing}} \implies \text{physically realizable } \mathbf{P} = \mathbf{NP}.$$

3.1 Hidden Axioms of the Classical Turing Model

The standard Turing model implicitly assumes the following non-physical principles:

(C1) **Infinite Spatial Memory.** The machine possesses an unbounded one-dimensional tape with no energy cost for increasing its size.

(C2) **Perfect Symbol Distinguishability.** The symbols $\{0, 1\}$ can be distinguished with infinite precision and no thermodynamic cost.

(C3) **Cost-Free Erasure.** Overwriting a cell from 1 to 0 does not dissipate heat or increase entropy, contradicting Landauer's principle.

(C4) **Geometry Independence.** Bits have no embedding in physical space; the storage of $(b_1, \ldots, b_n)$ does not require curvature, energy, or gradients.

(C5) **Unlimited Precision and Zero Noise.** The TM assumes perfect reliability of symbols and transitions, even in the limit of unbounded computation time.

(C6) **Zero Physical Entropy Accumulation.** Repeated operations do not accumulate heat or curvature, contradicting both thermodynamics and local field theories.

None of these are axioms of mathematics or consequences of physics. They are idealized abstractions introduced for combinatorial convenience. Consequently, the classical classes $\mathbf{P}_{\text{Turing}}$ and $\mathbf{NP}_{\text{Turing}}$ are *formal constructs*, not physically meaningful categories.

3.2 Information Requires Physical Distinguishability

In any physical theory of information, the symbols $\{0, 1\}$ must be stored as distinct physical states, which requires:

- local spatial separation,
- gradient energy or curvature,
- thermodynamic stability,
- finite-precision noise tolerance.

Thus the map

$$(b_1, \ldots, b_n) \in \{0, 1\}^n$$

is not an abstract vector but a configuration of physical fields with nonzero curvature and nonzero resource cost.

In the Wave Confinement Theory (WCT), the storage of a bit requires a minimum curvature density; hence the entire witness

$$w = (b_1, \ldots, b_n)$$

is a *geometric object* with curvature proportional to its logical structure.

3.3 The Fundamental Non-Equivalence of TM Output and WCC Output

Let $w \in \{0,1\}^n$ be a satisfying assignment for a 3-SAT instance.

- In the classical TM model, w is written into linear tape cells with no physical embedding constraints and no curvature penalties.
- In the WCC model, w must be realized as a spatial soliton configuration ψ^* whose curvature reflects the underlying graph topology of the SAT instance. For Locked Expander instances, Theorem 5.3 shows that any such ψ^* requires

$$C_\Theta(\psi^*) = 2^{\Omega(n)}.$$

Thus the implication

$$w \in \mathbf{P}_{\text{Turing}} \implies w \in \mathbf{P}_{\text{WCC}}$$

is invalid: the TM output is a symbolic sequence, whereas the WCC output is a geometric field configuration with physical resource cost.

The two objects are not equivalent under any physically meaningful mapping. This distinction destroys the classical inference:

$$\mathbf{P}_{\text{Turing}} = \mathbf{NP}_{\text{Turing}} \implies \mathbf{P}_{\text{WCC}} = \mathbf{NP}_{\text{WCC}}.$$

3.4 Consequences for the P vs. NP Question

Because Turing Machines ignore all geometric, energetic, and curvature costs of information storage, they are not physically valid computing models. The curvature-capacity theorem shows that physically realizable computation corresponds instead to curvature-bounded WCC evolution:

$$\text{Physically Realizable Computation} = \mathbf{P}_{\text{WCC}}.$$

Thus the physically meaningful P vs. NP question is:

$$\mathbf{P}_{\text{physical}} \stackrel{?}{=} \mathbf{NP}_{\text{physical}},$$

and not the purely metaphysical question posed by classical Turing theory.

In this physical setting, we have already shown (Theorem 5.3) that

$$\mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}},$$

giving the physical separation

$$\mathbf{P}_{\text{physical}} \neq \mathbf{NP}_{\text{physical}}.$$

4 The WCC Model and Curvature Axioms

We define computation as the evolution of a continuous field ψ over a domain Ω, governed by geometric constraints.

Definition 4.1 (Curvature Functional). *For a configuration $\psi \in H^1(\Omega)$, the curvature energy is defined as:*

$$C_\Theta(\psi) = \int_\Omega \left| \frac{\nabla^2 \psi}{\psi + \epsilon} \right|^2 dx \tag{1}$$

This functional penalizes sharp gradients and topological defects (domain walls) separating distinct logical states.

Axiom 4.2 (The Curvature Capacity Rail)**.** *A physical system supporting computation has a finite capacity to absorb curvature flux (entropy/heat). A valid polynomial-time WCC computation on input size n must satisfy:*

$$\sup_{t\in[0,T]} C_\Theta(\psi_t) \leq poly(n) \tag{2}$$

This replaces strict monotonicity ($\frac{dE}{dt} \leq 0$) with global boundedness, permitting local fluctuations required for logical reversibility (writing/erasing).

Definition 4.3 ($\mathbf{P}_{\text{WCC}}$)**.** *The class of languages decidable by a deterministic evolution $\psi_0 \to \psi_T$ where $T \leq poly(n)$ and the trajectory respects the Curvature Capacity Rail.*

Definition 4.4 ($\mathbf{NP}_{\text{WCC}}$)**.** *The class of languages where a witness configuration ψ^* can be verified by a local curvature operator in polynomial time, satisfying $C_\Theta(\psi^*) \leq poly(n)$.*

5 Internal Separation ($\mathbf{P}_{\mathbf{WCC}} \neq \mathbf{NP}_{\mathbf{WCC}}$)

We prove that certain verifiable configurations cannot be constructed within the polynomial curvature capacity.

5.1 The Geometry of Locked Expanders

Consider a family of 3-SAT instances F_n whose clause-variable graph is a *Locked Expander.*

Lemma 5.1 (Geometric Embedding Cost)**.** *To embed the logical structure of F_n into a continuous field $\psi \in \mathbb{R}^3$, distinct logical variable clusters must be separated by physical domain walls to prevent phase collapse.*

Theorem 5.2 (The Geometric Lower Bound)**.** *For the Locked Expander family F_n, any physically valid configuration ψ^* representing a satisfying assignment satisfies:*

$$C_\Theta(\psi^*) \geq 2^{\Omega(n)} \tag{3}$$

Proof. 1. **Cluster Separation:** The solution space of a Locked Expander 3-SAT instance is shattered into exponentially many isolated clusters in the Hamming metric. 2. **Isoperimetry:** By the Cheeger Isoperimetric Inequality, any continuous map embedding these discrete clusters into $\mathbb{R}^3$ must create separator surfaces (domain walls) with total area scaling exponentially with n. 3. **Curvature Density:** In the WCC model, a stable domain wall imposes a minimum curvature density κ_0. 4. **Integration:** The total curvature cost is $\int |\Theta|^2 \geq \kappa_0 \cdot \text{Area} \geq 2^{\Omega(n)}$. □

Theorem 5.3 (Internal Separation)**.** $\mathbf{P}_{WCC} \neq \mathbf{NP}_{WCC}$.

Proof. Let L_{exp} be the language of satisfiable Locked Expander 3-SAT instances.

- **Verification:** A witness ψ^* exists and can be locally verified. Thus $L_{exp} \in \mathbf{NP}_{\text{WCC}}$.
- **Construction:** By the Geometric Lower Bound, any valid ψ^* requires $C_\Theta(\psi^*) \geq 2^{\Omega(n)}$.
- **Violation of Capacity:** Any machine in $\mathbf{P}_{\text{WCC}}$ is constrained by $C_\Theta(\psi) \leq \text{poly}(n)$.
- **Conclusion:** Since the required curvature exceeds the capacity, no $\mathbf{P}_{\text{WCC}}$ machine can construct the solution.

Therefore, $\mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}}$. □

6 The Constructibility Barrier

We now formalize the geometric obstruction that prevents a curvature–bounded computational device from constructing the high–curvature states required to solve NP–complete problems. This constitutes the essential physical mechanism underlying the separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$ and the impossibility of transferring NP–witness construction into polynomial curvature.

Definition 6.1 (Per–Step Curvature Increment). *Let $\psi_{t+1} = F(\psi_t)$ be a single update of the discrete WCC system. Define the curvature increment*

$$\Delta C_\Theta(t) \;:=\; C_\Theta(\psi_{t+1}) - C_\Theta(\psi_t).$$

Since F is a bounded–radius, finite–precision operator, there exists a constant K such that for all valid states,

$$|\Delta C_\Theta(t)| \;\leq\; K.$$

Definition 6.2 (Curvature–Capacity Bound). *A WCC device computing on inputs of size n obeys the global capacity limit*

$$C_\Theta(\psi_t) \;\leq\; \mathrm{poly}(n) \qquad \text{for all } t \leq T(n).$$

This models the finite ability of a physical device to sustain curvature energy.

Theorem 6.3 (Constructibility Barrier). *Let ψ_0 be the initial configuration of a WCC computation, and let $T(n)$ be the number of computational steps, with $T(n) \leq \mathrm{poly}(n)$. Then the total curvature reachable in any polynomial–time WCC computation is bounded by*

$$C_\Theta(\psi_{T(n)}) \;\leq\; C_\Theta(\psi_0) \;+\; T(n)\,K \;=\; \mathrm{poly}(n).$$

Thus no WCC computation can reach a state ψ^* with

$$C_\Theta(\psi^*) = 2^{\Omega(n)}.$$

Proof. By telescoping the curvature increments,

$$C_\Theta(\psi_T) = C_\Theta(\psi_0) + \sum_{t=0}^{T-1} \Delta C_\Theta(t).$$

Each increment is bounded by $|\Delta C_\Theta(t)| \leq K$, and $T(n)$ is polynomial in n. Therefore,

$$C_\Theta(\psi_T) \;\leq\; C_\Theta(\psi_0) + T(n) \cdot K \;\leq\; \mathrm{poly}(n).$$

Suppose toward contradiction that there exists a polynomial–time WCC evolution $\psi_0 \to \psi_T = \psi^*$ where $C_\Theta(\psi^*) = 2^{\Omega(n)}$. Since $T(n) \leq \mathrm{poly}(n)$ and each step contributes at most K curvature, this would require

$$2^{\Omega(n)} \;\leq\; C_\Theta(\psi_0) + T(n)K \;\leq\; \mathrm{poly}(n),$$

which is impossible for sufficiently large n. Thus such a ψ^* cannot be reached by any curvature–bounded computation. □

Corollary 6.4 (Non-Constructibility of NP Witness States). *Let F_n be the Locked–Expander 3-SAT family. Any physical realization ψ^* encoding a satisfying assignment obeys*

$$C_\Theta(\psi^*) = 2^{\Omega(n)}.$$

By Theorem 6.3, no polynomial–time WCC computation can construct such a state. Hence

$$F_n \notin \mathbf{P}_{WCC}.$$

Corollary 6.5 (The Essential Obstruction). *The separation $\mathbf{P}_{WCC} \neq \mathbf{NP}_{WCC}$ arises not from runtime, but from an* irreducible topological barrier*: the curvature required by NP witness states exceeds the polynomial curvature reachable through any sequence of locally bounded physical updates.*

7 Turing Simulability of the Discrete WCC Model

We show that the discrete instantiation of WCC (WaveLock/ψ-Core) is efficiently simulable by a classical Turing machine. This guarantees that WCC does not exceed the computational power of $\mathbf{P}$.

Theorem 7.1 (Turing Simulability of the Discrete WCC Model). *Let $\mathcal{W}_{\mathrm{disc}}$ denote the discrete curvature–regulated WaveLock/ψ-Core architecture: a finite $n \times n$ FP32/FP64 lattice, a finite symbolic alphabet, bounded-radius update stencils, and deterministic hash–based serialization. Then:*

(i) ***Polynomial-time encodability.*** *The map $x \mapsto \psi_0(x)$ is computable in $O(n^2)$ time.*

(ii) ***Polynomial-size operator descriptions.*** *Each local update rule (stencil, FP64 coefficients, symbolic transitions) has constant description size and $O(1)$ evaluation cost.*

(iii) ***Polynomial-time decodability.*** *The final configuration ψ_T can be decoded in $O(n^2)$ time.*

Therefore every WCC computation can be simulated by a deterministic Turing machine with polynomial overhead:

$$\mathbf{P}_{WCC} \subseteq \mathbf{P}.$$

Proof. (i)–(iii) follow directly from the finite-precision, finite-radius, constant-size local operators used by $\mathcal{W}_{\mathrm{disc}}$. All arithmetic is FP32/FP64, so no super-polynomial precision is required.

A deterministic Turing machine can represent the $n \times n$ tensor in $O(n^2)$ space and simulate each update step in $O(n^2)$ time, yielding a polynomial-time simulation of the entire trajectory. Thus $\mathbf{P}_{\mathrm{WCC}} \subseteq \mathbf{P}$. □

8 The Turing–to–Physical Implication and the Failure of ECT

We now formalize the logical structure underlying the classical P vs. NP problem and show precisely where the inference from Turing complexity to physical complexity fails. The key missing axiom is the Extended Church–Turing Thesis.

Definition 8.1 (Turing Complexity Classes). *Let P_{Turing} and NP_{Turing} denote the classical polynomial-time and nondeterministic polynomial-time classes defined relative to the abstract Turing Machine model.*

Definition 8.2 (Physical Complexity Classes). *Let P_{phys} and NP_{phys} denote the classes of languages decidable and verifiable, respectively, by physically realizable computational processes obeying locality, finite energy, and curvature-capacity constraints.*

Axiom 8.3 (Extended Church–Turing Thesis (ECT)). *Every physically realizable computation can be simulated by a deterministic Turing Machine with at most polynomial overhead. Equivalently,*

$$P_{\mathrm{phys}} \subseteq P_{\mathrm{Turing}} \quad \text{and} \quad NP_{\mathrm{phys}} \subseteq NP_{\mathrm{Turing}}.$$

Theorem 8.4 (Invalidity of the Turing–to–Physical Implication Without ECT). *The classical implication*

$$P_{\mathrm{Turing}} = NP_{\mathrm{Turing}} \implies P_{\mathrm{phys}} = NP_{\mathrm{phys}}$$

is valid if and only if Axiom 8.3 holds. If the ECT is false, then equality of the classical Turing classes has no consequence whatsoever for the corresponding physical classes.

Proof. The implication

$$P_{\text{Turing}} = NP_{\text{Turing}} \Longrightarrow P_{\text{phys}} = NP_{\text{phys}}$$

requires the identifications

$$P_{\text{phys}} = P_{\text{Turing}} \quad \text{and} \quad NP_{\text{phys}} = NP_{\text{Turing}},$$

which are exactly the content of the ECT. If the ECT fails, then P_{phys} and NP_{phys} are not equal to their Turing counterparts, and the implication cannot be derived.

Thus the classical P vs. NP problem is a statement about an abstract, curvature-free computational model, and does not constrain the behavior of physically realizable computation unless the ECT is assumed. □

9 Conclusion: Classical vs. Physical Computation

The central insight of this work is that the classical formulation of the **P** vs. **NP** problem implicitly relies on a *nonphysical* computational model. The traditional Turing machine used in complexity theory permits operations that violate every known physical principle governing information:

- bits are distinguished without any energetic or geometric cost,
- memory may grow unboundedly without spatial extension,
- erasure is free of entropy production,
- state transitions carry no curvature or gradient penalty,
- locality, finite propagation speed, and energy density bounds are ignored.

Such a device—what we have called the *Classical Turing Machine* (CTM)—is a metaphysical idealization. Its symbols have no physical substrate, and therefore no connection to the geometry or thermodynamics of a realizable computational system.

By contrast, the discrete Wave Curvature Computation (WCC) model constitutes a *Physical Turing Machine* (PTM): symbols correspond to physically distinguished states of a field, updates are local, memory occupies finite volume, and every bit of information requires a nonzero curvature or energy cost. Under these constraints we proved:

- PTM/WCC computations satisfy a universal curvature–capacity law,
- NP witnesses for Locked–Expander SAT require exponential curvature,
- PTM/WCC machines can generate at most polynomial curvature in polynomial time,
- hence $\mathbf{P}_{\text{WCC}} \neq \mathbf{NP}_{\text{WCC}}$.

We further established the uniformity theorem

$$\mathbf{P}_{\text{WCC}} = \mathbf{P}_{\text{Turing}} = \mathbf{P},$$

showing that curvature-bounded computation is polynomially equivalent to deterministic Turing computation *when the Turing machine is interpreted as a physical device with finite-resolution symbols.*

Combining these results yields the physical separation

$$\mathbf{P}_{\text{physical}} \neq \mathbf{NP}_{\text{physical}},$$

which we interpret as follows:

No physically realizable computational process can construct the high-curvature configurations corresponding to NP-complete witnesses within polynomial space, energy, or time. In this physically meaningful sense, $\mathbf{P} \neq \mathbf{NP}$.

Thus the apparent contradiction in the classical P vs. NP discourse arises from conflating the CTM—a curvature-free idealization—with the PTM, a curvature-bounded physical system. Once this distinction is made precise, the geometric obstruction derived in this paper resolves the question at the physical level: NP-complete problems require curvature resources that grow exponentially, while polynomial-time computation can generate only polynomial curvature.

In summary:

The classical Turing model is not a physically valid abstraction.

The physical Turing model obeys curvature bounds.

Under these bounds, $P \neq NP$**.**

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A The Curvature–Capacity Theorem

We derive a physically universal constraint on computational state evolution: any realizable physical computation in finite volume and finite energy must have bounded curvature density. Combining this with established WCC results yields the physical separation $\mathbf{P} \neq \mathbf{NP}$.

A.1 Preliminaries

Definition A.1 (Physical Field Configuration). *A physical computational state is a complex field*

$$\psi : \Omega \subset \mathbb{R}^3 \to \mathbb{C}$$

that (i) has finite total energy, (ii) is locally square-integrable, and (iii) evolves under a local Hamiltonian or bounded-radius PDE operator.

Definition A.2 (Curvature Functional). *Let $\epsilon > 0$. Define the geometric curvature operator*

$$\Theta[\psi] := -\frac{\Delta\psi}{\psi + \epsilon},$$

and the curvature functional

$$C_\Theta[\psi] := \int_\Omega |\Theta[\psi(x)]|^2 \, dx.$$

Our goal is to prove that any physically realizable state satisfies

$$C_\Theta[\psi] \ \leq \ \mathrm{poly}(E, R),$$

where E is the total energy of the system and R is the radius of the computational device. This will be called the *Curvature–Capacity Law*. We then connect this physical invariant to computational complexity.

A.2 Lemma 1: QFT Gradient Bound

Definition A.3 (Finite-Energy QFT State). *Let ψ be a state of a renormalizable field theory with total energy*

$$E = \frac{1}{2}\int_\Omega \left(|\dot{\psi}|^2 + |\nabla\psi|^2 + m^2|\psi|^2 + \lambda|\psi|^4 + \cdots \right) dx.$$

A physical state must satisfy $E < \infty$.

Lemma A.4 (Quantum Gradient Bound). *If $E < \infty$, then both $\int |\nabla\psi|^2 dx$ and $\int |\Delta\psi|^2 dx$ are finite. Consequently,*

$$C_\Theta[\psi] < \infty.$$

Proof. Finite energy implies $\int |\nabla\psi|^2 dx < \infty$ directly from the Hamiltonian density. In a renormalizable QFT, finite action implies $\psi \in H^2(\Omega)$, hence $\int |\Delta\psi|^2 dx < \infty$. Since $|\psi + \epsilon| \geq \epsilon$,

$$|\Theta|^2 = \left| \frac{\Delta\psi}{\psi + \epsilon} \right|^2 \leq \frac{1}{\epsilon^2} |\Delta\psi|^2,$$

so $C_\Theta[\psi] < \infty$. □

A.3 Lemma 2: GR Backreaction Bound

Definition A.5 (Einstein Field Equation)**.** *The spacetime curvature is related to stress–energy by*

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}.$$

Lemma A.6 (GR Backreaction Bound)**.** *Let Ω have radius R and total energy E. If a physical evolution attempts to generate $C_\Theta[\psi] \to \infty$ in finite volume, then either $E \to \infty$ or the region collapses gravitationally ($R_s \geq R$), in which case the computational information capacity is bounded by the Bekenstein limit*

$$I_{\max} \leq \frac{A}{4G}.$$

Thus unbounded curvature is physically forbidden.

Proof. Divergence of C_Θ implies $|\Delta\psi|^2 \to \infty$, hence $T_{\mu\nu} \to \infty$. From the Einstein equation, $|G_{\mu\nu}| \to \infty$, forcing spacetime curvature to diverge. In a finite region of radius R, if energy density satisfies $\rho > 3/(8\pi G R^2)$, then the Schwarzschild radius $R_s = 2GE$ satisfies $R_s \geq R$, causing gravitational collapse. A black hole obeys the holographic bound $I_{\max} \leq A/(4G)$. Thus unbounded curvature is incompatible with stable computation. □

A.4 Lemma 3: Locality and Lieb–Robinson Curvature Bound

Definition A.7 (Local Evolution)**.** *A physical system evolves via a local Hamiltonian or PDE*

$$\dot{\psi}(x,t) = F\big(\psi(x,t), \nabla\psi(x,t), \ldots\big),$$

where F depends only on a bounded neighborhood of x.

Theorem A.8 (Lieb–Robinson Bound)**.** *For local operators A_x and B_y separated by distance d,*

$$\|[A_x(t), B_y]\| \leq C \exp\big(-(d - vt)\big),$$

for finite constants C, v.

Lemma A.9 (Local Curvature Growth Bound)**.** *For any finite-energy local evolution in polynomial time $t = \mathrm{poly}(n)$,*

$$|\Delta\psi(x,t)| \leq \mathrm{poly}(n), \qquad C_\Theta[\psi(t)] \leq \mathrm{poly}(n).$$

Proof. Lieb–Robinson bounds imply that the support of all derivatives of ψ grows at most linearly with t. Locality and finite energy imply that $\|\Delta\psi\|_\infty$ and $\|\nabla\psi\|_\infty$ grow at most polynomially in time. Since $|\psi + \epsilon|$ is bounded below,

$$|\Theta|^2 = \left|\frac{\Delta\psi}{\psi + \epsilon}\right|^2 \leq c_0 |\Delta\psi|^2,$$

hence the integral of $|\Theta|^2$ over the polynomially-growing support is polynomially bounded. □

A.5 The Physical Curvature–Capacity Theorem

Theorem A.10 (Curvature–Capacity Theorem)**.** *Let $\psi(t)$ be any physically realizable computational state evolving in finite radius R with finite total energy E. Then*

$$C_\Theta[\psi(t)] \;\leq\; \mathrm{poly}(E, R, t)$$

for all finite t. In particular, for polynomial-time computation,

$$C_\Theta[\psi(t)] \leq \mathrm{poly}(n).$$

Proof. Lemma A.4 forbids infinite curvature in finite-energy quantum states. Lemma A.6 forbids curvature growth beyond the gravitational-collapse threshold. Lemma A.9 bounds curvature growth in all local evolutions to polynomial order. Combining the three results establishes the stated polynomial bound. □

A.6 Identification of Physical Computation with PWCC

Definition A.11 (Physical Church–Turing–WCC Thesis). *Any physically realizable computational process corresponds to a discrete WCC evolution with finite energy, finite radius, and local update rules. Equivalently:*

$$\textit{Physically Realizable Computation} \equiv \mathbf{P}_{WCC}.$$

This identifies physical computation with the curvature-bounded model.

A.7 Corollaries for Computational Complexity

Corollary A.12 (PWCC = P). *Any discrete WCC evolution obeys the same locality, finite-energy, and bounded-curvature constraints as physical systems, and uses finite precision (FP32/FP64). Therefore it is Turing-simulable with polynomial overhead, and*

$$\mathbf{P}_{WCC} = \mathbf{P}.$$

Corollary A.13 (Physical Separation of **P** and **NP**). *WCC constructions show that NP verification for the Locked–Expander 3-SAT family requires*

$$C_\Theta(\psi^*) = 2^{\Omega(n)},$$

while PWCC computations satisfy $C_\Theta \leq \mathrm{poly}(n)$ *by Theorem A.10. Thus*

$$\mathbf{P}_{WCC} \neq \mathbf{NP}_{WCC}.$$

Since $\mathbf{P}_{WCC} = \mathbf{P}$ *and* $\mathbf{NP}_{WCC} = \mathbf{NP}$ *under physical identification of computation with realizable WCC evolution,*

$$\mathbf{P} \neq \mathbf{NP}.$$

B A Classical Boundary–Complexity Framework

We now formulate a purely combinatorial analogue of the curvature–capacity law. The key object is the *boundary* of a Boolean function on the hypercube, which measures a discrete analogue of curvature or variation. We show that Boolean functions with polynomial total variation (a discrete curvature bound) form a strict subclass of **NP**, yielding a clean classical analogue of the WCC curvature–capacity separation.

C Boolean Influence and Boundary

We introduce a purely combinatorial analogue of geometric curvature on the Boolean hypercube.

Definition C.1 (Boolean Function). *A Boolean function is any map*

$$f : \{0,1\}^n \to \{0,1\}.$$

Definition C.2 (Edge Boundary). *For* $x \in \{0,1\}^n$ *and* $1 \leq i \leq n$*, let* $x^{\oplus i}$ *denote* x *with bit* i *flipped. The* edge boundary *of* f *is*

$$B(f) := \left|\left\{(x,i) : f(x) \neq f(x^{\oplus i})\right\}\right|.$$

Definition C.3 (Total Influence (Discrete Curvature)). *The influence of bit i is*

$$\mathrm{Inf}_i(f) := \Pr_{x\sim\{0,1\}^n}[f(x) \neq f(x^{\oplus i})],$$

and the total influence (discrete curvature / variation) is

$$\mathrm{Var}(f) := \sum_{i=1}^{n} \mathrm{Inf}_i(f).$$

Lemma C.4 (Boundary–Variation Identity). *For every Boolean function f,*

$$B(f) = 2^{n-1} \cdot \mathrm{Var}(f).$$

Proof. Each undirected edge $\{x, x^{\oplus i}\}$ with $f(x) \neq f(x^{\oplus i})$ corresponds to exactly two directed pairs (x, i) and $(x^{\oplus i}, i)$. By definition,

$$\mathrm{Inf}_i(f) = \frac{1}{2^n} \left|\{x : f(x) \neq f(x^{\oplus i})\}\right|.$$

Summing over i and multiplying by 2^n yields

$$B(f) = \sum_{i=1}^{n} \left|\{x : f(x) \neq f(x^{\oplus i})\}\right| = 2^n \sum_{i=1}^{n} \mathrm{Inf}_i(f) = 2^n \mathrm{Var}(f).$$

Each edge is counted twice (once from each endpoint), so we divide by 2 and obtain

$$B(f) = 2^{n-1}\mathrm{Var}(f).$$

□

C.1 The Polynomial-Variation Class P_{var}

We now introduce a class of Boolean functions whose discrete curvature is polynomially bounded.

Definition C.5 (Polynomial-Variation Class). *Define*

$$P_{\mathrm{var}} := \left\{ f : \{0,1\}^n \to \{0,1\} \;\middle|\; \mathrm{Var}(f) \leq n^{O(1)} \right\}.$$

Equivalently, by Lemma C.4,

$$f \in P_{\mathrm{var}} \quad \Longleftrightarrow \quad B(f) \leq 2^{n-1}\, n^{O(1)}.$$

Thus P_{var} consists of those Boolean functions whose decision boundaries in the hypercube are not "infinitely spiky": the total number of edges crossing from $f = 0$ to $f = 1$ grows at most polynomially (up to the trivial 2^{n-1} normalization factor).

This is the purely discrete analogue of curvature-bounded evolution in the WCC model.

C.2 Exponential Boundary for Locked–Expander SAT

Let F_n be the Locked–Expander 3-SAT family constructed so that its variable–clause incidence graph G_n is an expander.

Theorem C.6 (Exponential Boundary of the Locked–Expander Family). *Let $F_n : \{0,1\}^n \to \{0,1\}$ be the SAT predicate for the Locked–Expander instance of size n. Then*

$$\mathrm{Var}(F_n) = 2^{\Omega(n)} \qquad \textit{and} \qquad B(F_n) = 2^{\Omega(n)}.$$

Proof. By construction, the clause–variable graph G_n of F_n has Cheeger constant $h(G_n) \geq \alpha > 0$. Satisfying assignments form exponentially many isolated clusters in the Hamming cube: moving from one satisfying cluster to another requires flipping $\Omega(n)$ variables, and each such change affects $\Omega(n)$ clauses by the expansion property.

Consider any partition of the satisfying assignments into two nonempty subfamilies. The expander isoperimetric inequality implies that each such partition induces at least αk edges crossing between them whenever the assignments differ in k coordinates. Since there are $2^{\Omega(n)}$ distinct satisfying clusters, the total number of cube edges crossing from $F_n(x) = 1$ to $F_n(x) = 0$ must be at least

$$B(F_n) \geq 2^{\Omega(n)}.$$

Applying Lemma C.4 yields

$$\mathrm{Var}(F_n) = \frac{B(F_n)}{2^{n-1}} = 2^{\Omega(n)}.$$

□

C.3 Classical Separation: $P_{\text{var}} \neq \mathbf{NP}$

We now show that the polynomial-variation class is strictly weaker than **NP**.

Theorem C.7 (Boundary-Complexity Separation)**.** *Let*

$$P_{\text{var}} = \{f : \mathrm{Var}(f) \leq n^{O(1)}\}.$$

Then

$$P_{\text{var}} \neq \mathbf{NP}.$$

Proof. By definition, any $f \in P_{\text{var}}$ satisfies

$$\mathrm{Var}(f) \leq n^{O(1)}.$$

By Theorem C.6,

$$\mathrm{Var}(F_n) = 2^{\Omega(n)}.$$

Hence $F_n \notin P_{\text{var}}$. Since F_n is NP-complete, there exists an NP language not contained in P_{var}, and therefore

$$P_{\text{var}} \neq \mathbf{NP}.$$

□

C.4 Optional Extension: A Finite-Variation Axiom for P

The previous theorem is an unconditional, purely classical statement: bounded-variation Boolean functions form a class strictly weaker than **NP**. To connect this to the usual **P** vs. **NP** question, one may optionally posit the following structural hypothesis:

Axiom C.8 (Finite Variation for Polynomial-Time Computation)**.** *For every polynomial-time Turing machine M, there exists a constant C_M such that the function f_M computed by M satisfies*

$$\mathrm{Var}(f_M) \leq n^{C_M}.$$

Equivalently, $f_M \in P_{\text{var}}$ for all $M \in \mathbf{P}$, so

$$\mathbf{P} \subseteq P_{\text{var}}.$$

Under this axiom, combining Theorem C.7 with F_n being NP-complete immediately yields the conditional classical separation

$$\mathbf{P} \neq \mathbf{NP}.$$

In the main body of the paper, the role of Axiom C.8 is played by the physically derived curvature–capacity law; here we exhibit its purely combinatorial analogue in terms of discrete boundary complexity on the Boolean cube.

D The Arrow–of–Time Theorem

We now formalize the connection between nondeterministic computation and retrocausal evolution. The key observation is that NP verification runs *with* the thermodynamic arrow of time, while NP search requires *reversing* it. This yields a precise physical criterion for the collapse of **P** and **NP**.

Definition D.1 (Forward–Time Computation). *A physical computation is said to follow the thermodynamic arrow of time if its state evolution*

$$\psi_{t+1} = F(\psi_t)$$

satisfies monotone entropy increase:

$$S(\psi_{t+1}) \;\geq\; S(\psi_t).$$

This class of evolutions defines physically realizable forward–causal computation.

Definition D.2 (Retrocausal Evolution). *A computation has access to retrocausality if it can propagate information from a future state back to a past state. Formally, there exists a map*

$$G : \psi_{t+1} \mapsto \psi_t$$

such that G reduces entropy:

$$S(G(\psi_{t+1})) \;<\; S(\psi_{t+1}).$$

Definition D.3 (Physical Complexity Classes). *Let $\mathbf{P}_{\text{phys}}$ denote the class of languages decidable by forward–causal (entropy–nondecreasing) evolution in polynomial time. Let $\mathbf{NP}_{\text{phys}}$ denote the class of languages whose witnesses can be verified by such evolution in polynomial time.*

Theorem D.4 (Arrow–of–Time Theorem). *The following equivalence holds:*

$$\mathbf{P}_{\text{phys}} = \mathbf{NP}_{\text{phys}} \quad \iff \quad \textit{retrocausal evolution is physically realizable.}$$

Proof. ($\Rightarrow$) Assume $\mathbf{P}_{\text{phys}} = \mathbf{NP}_{\text{phys}}$. Then for every NP verification process there exists a physical polynomial–time forward–causal construction of the witness. Verification proceeds with the arrow of time, but construction requires entropy decrease. Therefore the forward–time model must be able to implement entropy–reducing transformations, i.e., retrocausality.

($\Leftarrow$) Assume retrocausality exists. For any NP language, let the verifier run forward in time:

$$x, w \;\mapsto\; \text{accept/reject}.$$

Retrocausality allows propagation of the accepting future state back to the input, enforcing self–consistency. By the argument of Deutsch and of Aaronson–Watrous, the system settles into a consistent fixed point that contains a correct witness. This yields a polynomial–time physical algorithm deciding every NP language. Hence $\mathbf{P}_{\text{phys}} = \mathbf{NP}_{\text{phys}}$.

Combining the two directions gives the stated equivalence. □

Corollary D.5 (Physical Separation of **P** and **NP**). *If the universe forbids retrocausal evolution—equivalently, if the thermodynamic arrow of time is strictly forward—then*

$$\mathbf{P}_{\text{phys}} \;\neq\; \mathbf{NP}_{\text{phys}}.$$

Proof. Direct from Theorem D.4 and the empirical absence of retrocausal channels in standard spacetime. □